Ke Lin

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Education

College of William & Mary — **Sep. 2025 – Jun. 2030 (Expected)**
Ph.D. in Data Science — *Williamsburg, Virginia, US*
Advisor: Prof. Qingyun Wang

Tsinghua University — **Sep. 2022 – Jun. 2025**
M.Eng. in Software Engineering (with Honors) — *Beijing, China*
Advisor: Prof. Ping Luo

Tsinghua University — **Sep. 2018 – Jun. 2022**
Bachelor in Software Engineering — *Beijing, China*
Advisor: Prof. Lijie Wen

Experience

Tencent — **Apr. 2024 – Sep. 2024**
AI Research Intern — *Shenzhen, China*

- Developed a VLLM-based pipeline for evaluating the realism, reliability, and aesthetics of generated 3D scenes.
- Built a 2D rendering framework using Blender for generating views of 3D objects as inputs to VLLMs.
- Optimized the LLM agents to generate 3D layouts for the compositions of pre-defined 3D assets.
- Investigated the spatial reasoning capability of LLMs to produce precise 3D layouts utilizing arithmetic coordinates.

Momenta — **Jan. 2021 – Apr. 2021**
Backend Intern — *Beijing, China*

- Developed an automated service for scheduling autonomous driving model training tasks based on Kubernetes.
- Utilized Golang to reduce redundant resource consumption and estimated the approximate cost of training sessions.
- Processed the runtime logs of training tasks and stored them in AWS Cloud Storage for visualization.
- Automated the deployment of driving models on AWS and the updating of the K8s image from the upstream repository.

Publications

1. Yiyang Luo*, **Ke Lin***, Chao Gu*, Jiahui Hou, Lijie Wen and Ping Luo. "Lost in Overlap: Exploring Logit-based Watermark Collision in LLMs." In Findings of the Association for Computational Linguistics: **NAACL 2025**.
   - Proposed the concept of watermark collisions, where multiple watermarks are present simultaneously in the same text.
   - Analyzed the potential risks and the vulnerability of existing watermarking techniques.
2. Yiyang Luo*, **Ke Lin*** and Chao Gu*. "Context-Aware Indoor Point Cloud Object Generation through User Instructions." In Proceedings of the 32nd ACM International Conference on Multimedia (**ACM MM'24**).
   - Designed a GPT-aided data pipeline for paraphrasing the descriptive texts in the ReferIt3D dataset to generative ones.
   - Proposed an end-to-end multi-modal diffusion model for generating indoor 3D objects into specific scenes.
   - Introduced the visual grounding task to assess the quality of an augmented scene, along with other metrics.
3. **Ke Lin**, Yiyang Luo, Zijian Zhang and Ping Luo "Zero-shot Generative Linguistic Steganography." In Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics (**NAACL 2024**).
   - Presented a zero-shot approach for linguistic steganography based on in-context learning using samples of covertexts.
   - Improved both the binary coding process and the embedding process by differential coding and annealing penalty.
   - Designed several metrics and language evaluations to evaluate both the perceptual and statistical imperceptibility.
4. **Ke Lin**, Yasir Glani, and Ping Luo. "Low-Latency Privacy-Preserving Deep Learning Design via Secure MPC." In Proceedings of the IJCAI-24 Workshop on Artificial Intelligence Safety (**IJCAI-AISafety 2024**).
5. Chao Gu, **Ke Lin**, Yiyang Luo, Jiahui Hou, and Xiang-Yang Li. "ViRED: Prediction of Visual Relations in Engineering Drawings." In Proceedings of the 2024 19th Int. Conf. on Mobility, Sensing and Networking (**IEEE MSN 2024**).
   - Developed a relation prediction pipeline to differentiate between the circuit and tabular parts in engineering drawings.
   - Achieved a 10% enhancement in detection accuracy compared to prior studies on the electrical diagram dataset.
6. Glani Yasir, Ping Luo and **Ke Lin**. "MDASC: Advanced Dual-Layered Code Cloning Technique for Identifying Reused Malicious Code." International Conference on Software Engineering and Knowledge Engineering (2024).
7. **Ke Lin** and Ping Luo. "Skipping Scheme for Gate-hiding Garbled Circuits." arXiv preprint arXiv:2312.02514 (2023).
8. Glani Yasir, Ping Luo, **Ke Lin**, et al. "AyatDroid: A Lightweight Code Cloning Technique Using Different Static Features." 2023 IEEE 3rd International Conference on Software Engineering and Artificial Intelligence (SEAI). IEEE, 2023.

9. Yue-Xun Huang, Ming Li, **Ke Lin**, et al. "Classical-to-quantum transition in multimode nonlinear systems with strong photon-photon coupling." **Physical Review A** 105.4 (2022): 043707.

*Equal contribution

Projects

Farthest Point Sampling Library | *Python, Rust, C++* **Sep. 2023**

- Developed a high-performance farthest point sampling library `fpsample` for Numpy arrays.
- Achieved 100× faster than vanilla implementation in pure Numpy for simplified preprocessing of 3D point clouds.
- Published PyPI packages for easy use in x64 platforms to avoid multi-language compilations.

Technical Skills

Languages: Mandarin (native), Cantonese (native), English (fluent) | Python, Golang, Rust, C++, Java, ReactJS, SQL
Tools/Frameworks: PyTorch, Blender, Ubuntu, ArchLinux, PostgreSQL

Awards & Scholarships

- 2025 Lambda Research Grant
- 2025 THSS Outstanding Graduate with Honors
- 2024 Tsinghua-Ubiquant Scholarship